

Protein-family
atlas

Two candidate
inventories

Candidate
annotations

Biological context
for candidate selection



426 assemblies

nifHDK labels

nifH

nifD

nifK



MMseqs2
family
matrix

426 × 2,286

Genus-blocked
prediction

Training-fold filters

Figure 2

Highly Pure

981

families

High-purity
diazotrophy context

Model-Supported

476

families

Accessory
candidate families



Active-phase-up
proteomics



Genus-based
morphotype proxies



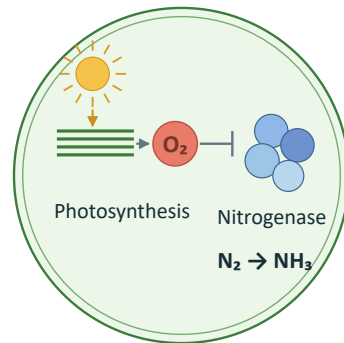
Condensate-
driver ranks



Redox and
electron flow



Fe-S and
cofactor handling



Carbon and
fixed-N
metabolism



Protein and RNA
maintenance